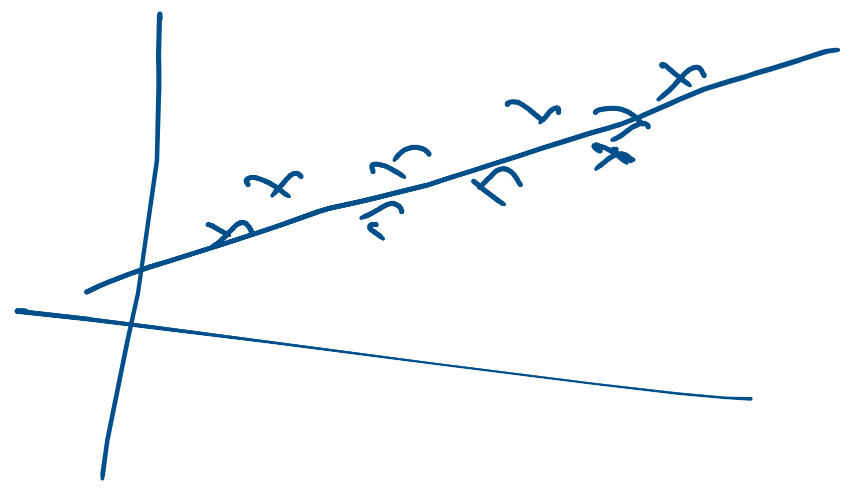
Linear Regression
- → Easy
- → Foundational Algo



Linear Regression → Supervised ML

LR
- Simple LR
- Multiple LR
- Polynomial LR

Simple LR

1 Input col | 1 output col

student_table

Iq	CGPA
9	3.85
5	2.70

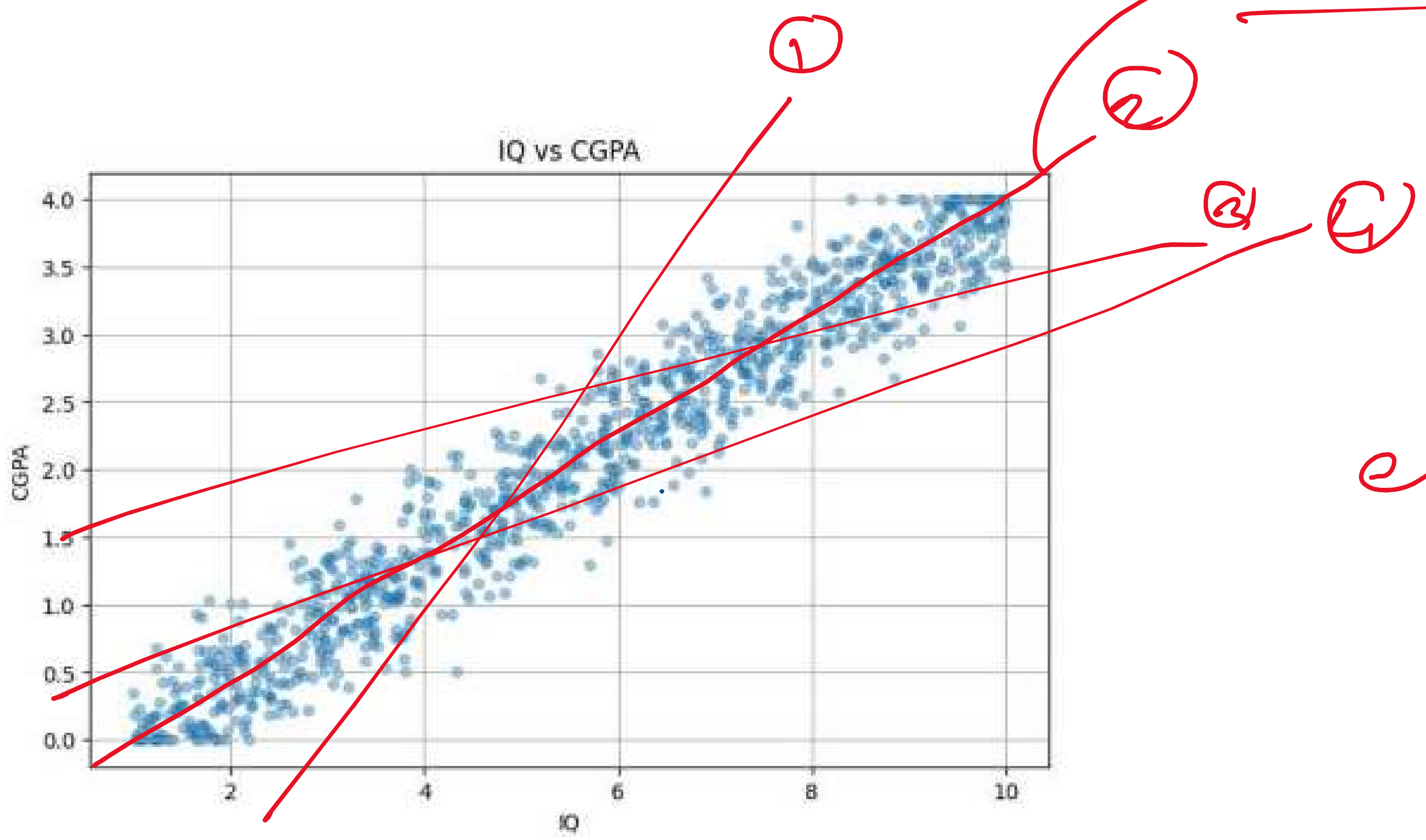
Input → Iq

CGPA → output

Input ← IQ	CGPA → Output
9	3.85
5	2.70
10	4
8.5	3.65

$y = mx + b$

Best fit line



$y = mx + b$

m, b

cost → Low $[m, b]$

IQ CGPA

$y = mx + b$

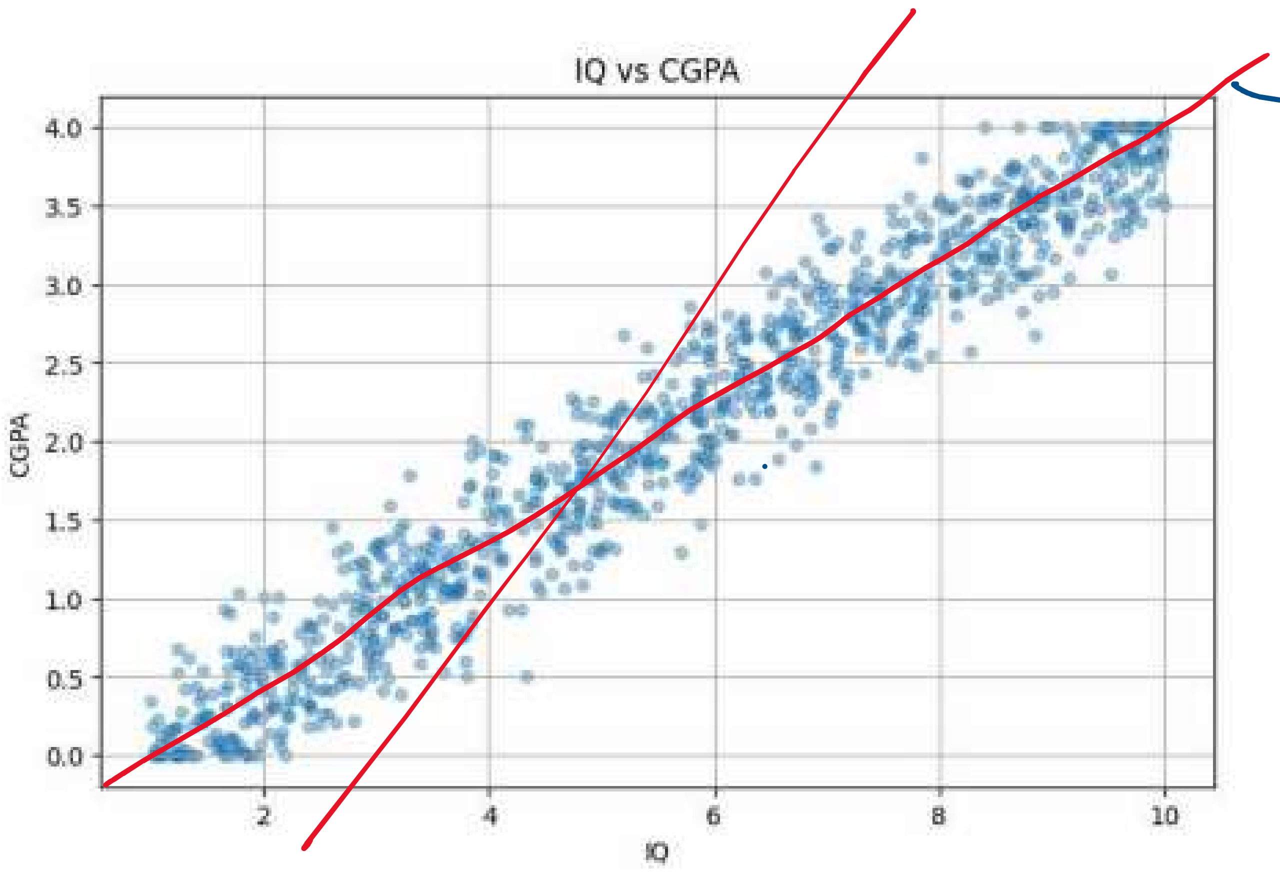
$y \to$ CGPA

$x \to$ IQ

$$\boxed{\begin{array}{l} m = 5 \\ b = 2 \end{array}}$$

$$cgpa = m \times Iq + b$$

$$= 5 \times \boxed{} + 2$$



$\rightarrow m, b$

$$y = mx + b$$

$$cgpa = m \times Iq + b$$

m = weightage

$m \uparrow$ = cgpa more depend on Iq

$m \downarrow$ = cgpa less depend on Iq

$$\boxed{b = 0}$$

$$cgpa = m \times Iq$$
$$= 0$$

Input

$Iq = 0$

Experience	Salary
1	25000
0	0

$$\text{salary} = m \times \text{Exp} + \boxed{b}$$ θ

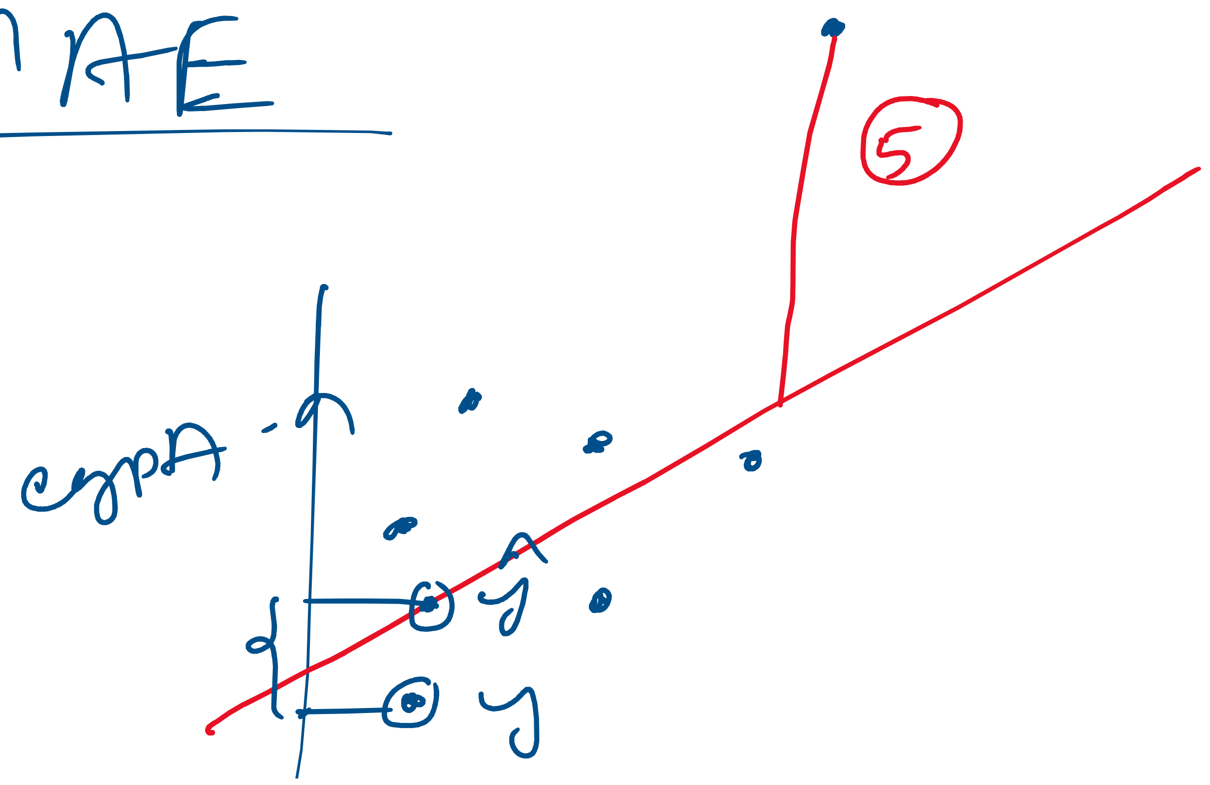
$$= 0 + b$$

10,000

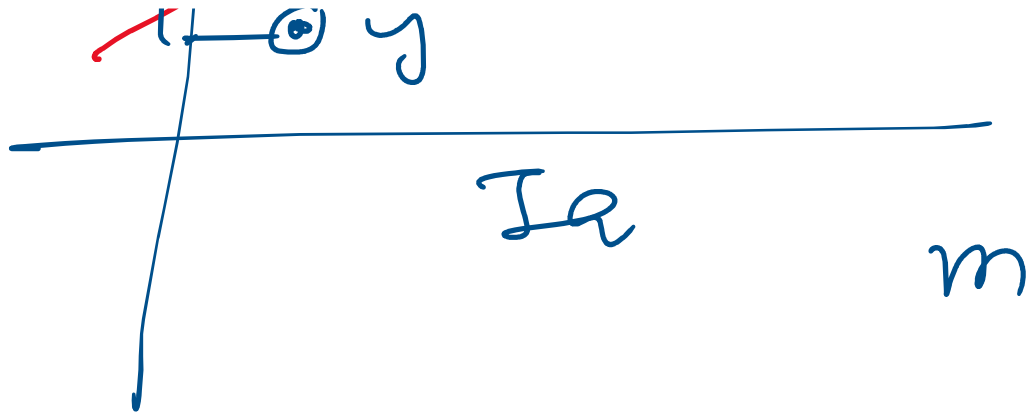
Regression Metrics

1) MAE → Mean absolute Error
2) MSE → " square "
3) RMSE → root mean square error
4) R_2 score

MAE



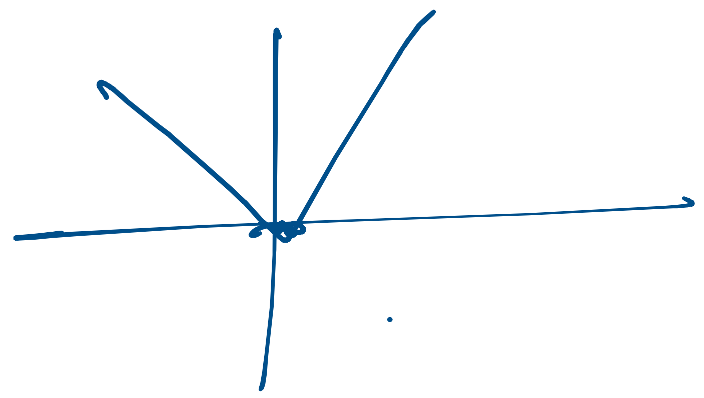
$$|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|$$



$$mae = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n} \qquad |0|$$

Advantage
- same unit
- Robust outlier

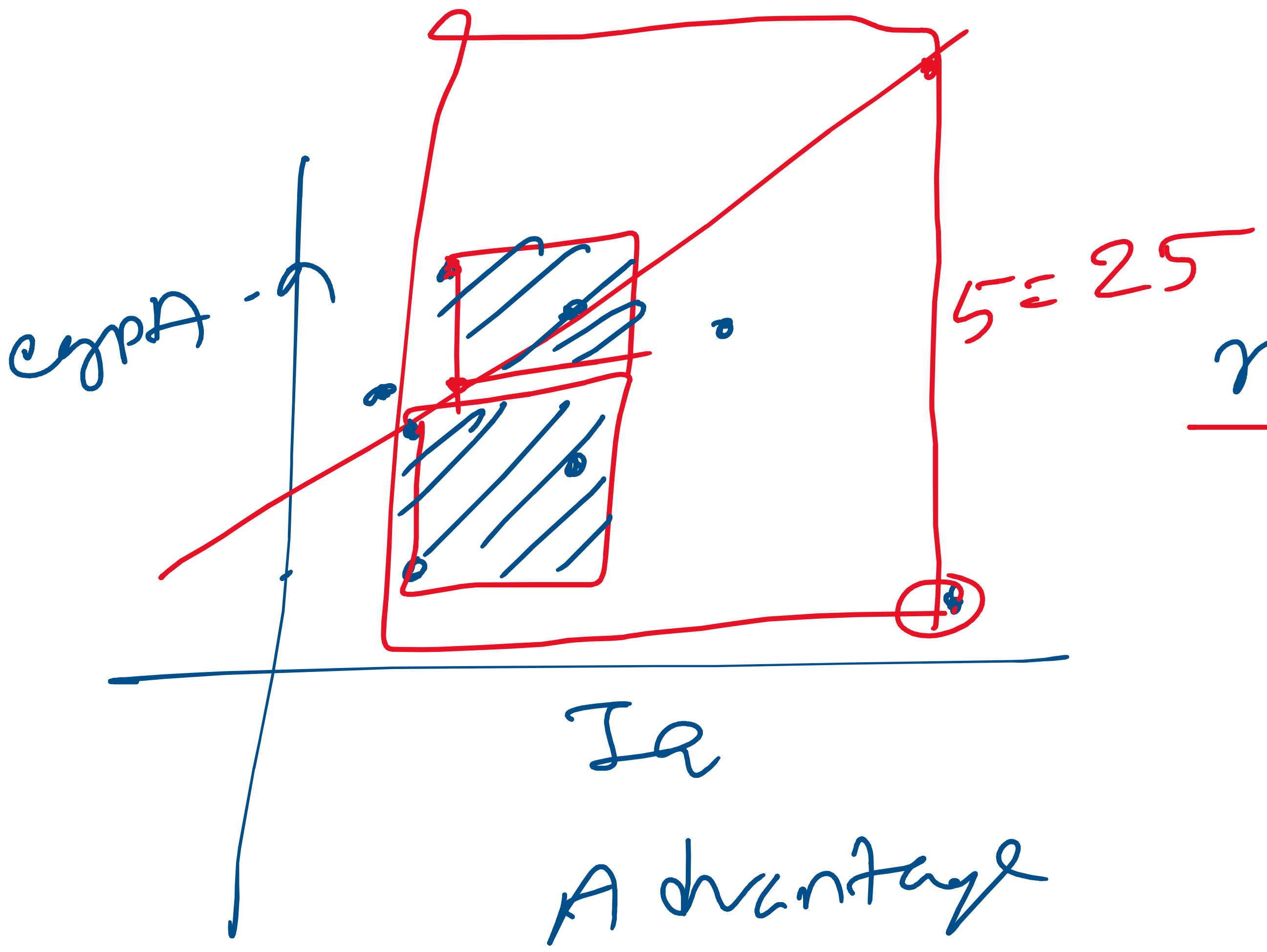
Disadvange



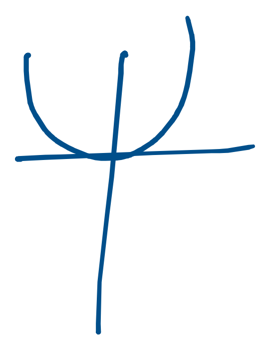
MSE

$$(0)^2$$

$$mae = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$



$$mse = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$



Advantage
- Loss function

Disadvantage
- → not same unit
- → Not Robust in outliers

Loss y $\hat{y}$

RMSE

$$= \sqrt{MSE} = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

Iq	cgpa	Pred_c	$y - \hat{y}$
10	4	3.8	0.20
8	3.5	3.4	0.10
9	3.8	3.75	0.05

w, y, cost

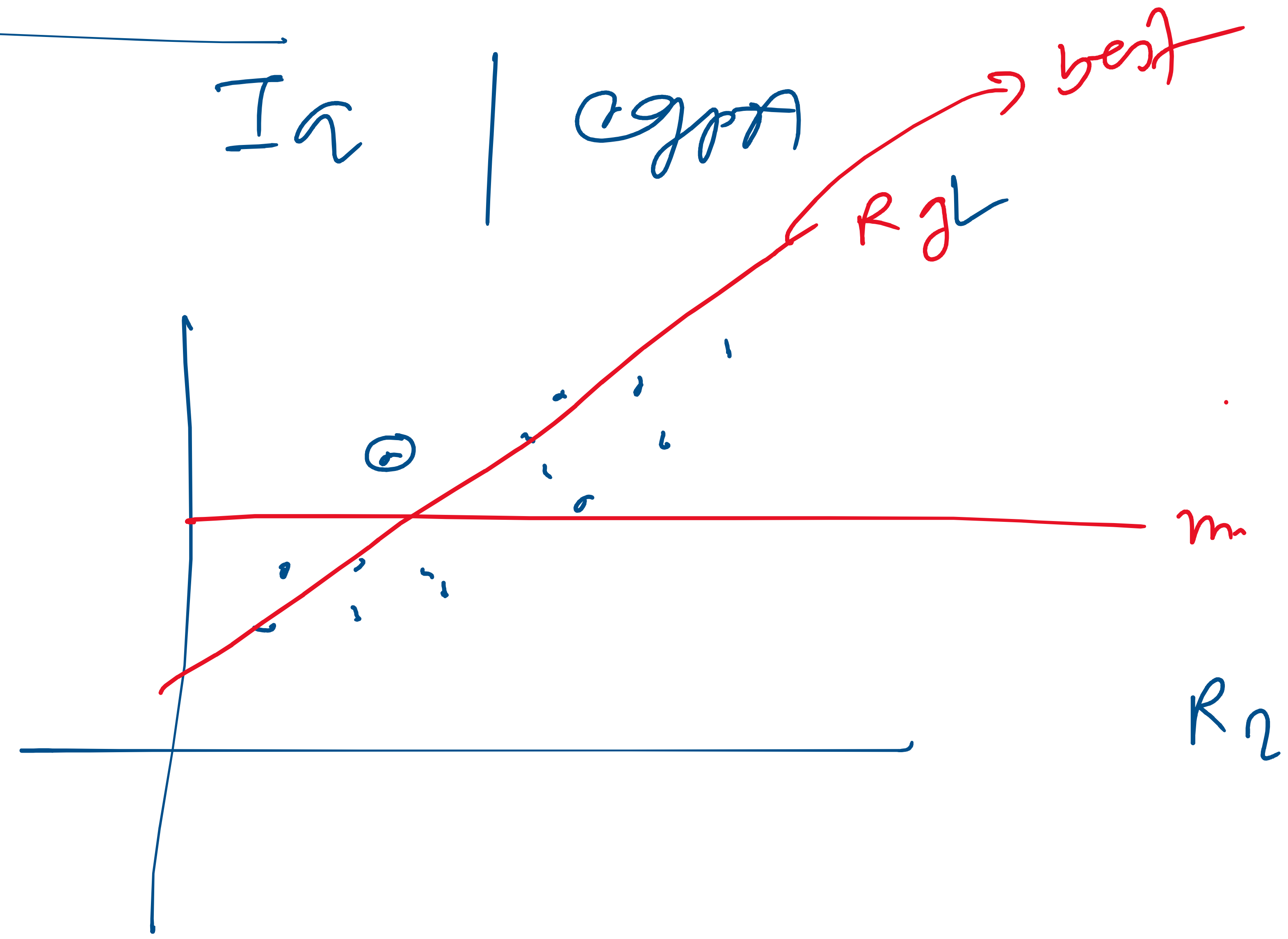
Advantage
- → same unit

Disadvantage
- → Not Robust in outliers

R_2 Score

100

Iq | cgpa



$\boxed{\text{output}}$

$$R_2 = 1 - \frac{SS_R}{SS_m}$$

SS_R = Sum of squared error in Regression Line

SS_m = Sum of squared error in mean Line

$$R_2 = 1 - \frac{\left[\sum_{i=1}^{n}(y_i-\hat{y}_i)^2\right]_{Rg}}{\left[\sum_{i=1}^{n}(y_i-\bar{y}_i)^2\right]_{mL}}$$

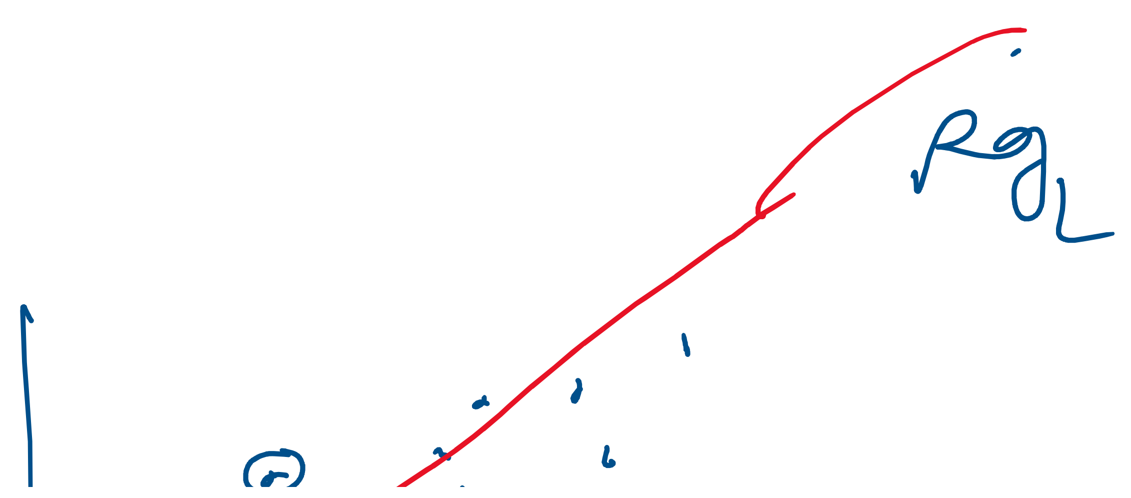
$$\boxed{R_2 = 0}$$

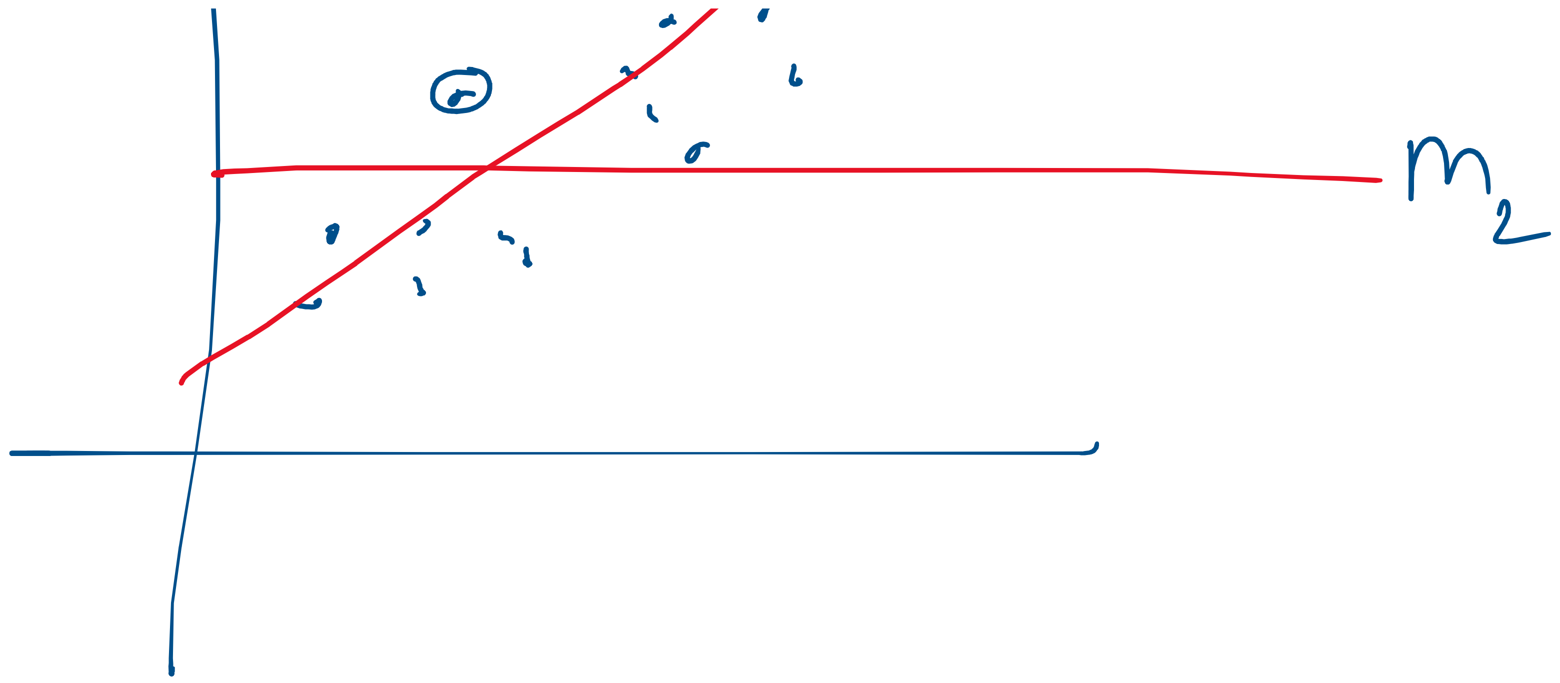
$$R_2 = 1 - \boxed{\overset{1}{\frac{\left[\sum_{i=1}^{n}(y_i-\hat{y}_i)^2\right]_{Rg}}{\left[\sum_{i=1}^{n}(y_i-\bar{y}_i)^2\right]_{mL}}}}$$

$$\overset{1}{\frac{Same}{Same}}$$

$$= 1 - 1$$
$$= 0$$

$R_2 = 0 \leftarrow$ wrong





O

$R_2 = 1$

$$R_2 = 1 - \boxed{\frac{\left[\sum_{i=1}^{n}(y_i - \hat{y}_i)^2\right]_{Rg}}{\left[\sum_{i=1}^{n}(y_i - \bar{y}_i)^2\right]_{mL}}}$$

$R_2 \rightarrow 1$ (খুব ভালো) $= 1 - \cancel{0}$

$= 1$

0.25

$\boxed{\begin{array}{l} 0.85 \\ ০.৯০ \end{array}}$

$R_2 = 0.85$

$\boxed{IQ}$ explain 85%

Variance

IQ	CGPA

R^2 = 0.85 মানে IQ CGPA কে বেশ ভালোভাবে explain করতে

Input

Ex | IQ | CGPA |

Output

Salary

R^2 = 0.85 মানে IQ CGPA কে বেশ ভালোভাবে explain করতে পারে, কিন্তু পুরোটা না।

$R^2 = 0.90$